PromptRGD: Prompt Learning with Relation-aware Gradient Denoising for Low-resource Relation Extraction

Zihang Xu
Department of Computer Science
Shanghai Jiao Tong University
Shanghai, China
zihang_x@sjtu.edu.cn

Luoyi Fu✉
Department of Computer Science
Shanghai Jiao Tong University
Shanghai, China
yiluofu@sjtu.edu.cn

Xinbing Wang
Department of Computer Science
Shanghai Jiao Tong University
Shanghai, China
xwang8@sjtu.edu.cn

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***Abstract*—Relation extraction is a fundamental task to construct a knowledge graph, which aims to find the relation categories between two entities from a string of text input sequences. Considering the fact that high-quality labeled data is rare, a recent trend is low-resource setting. Existing works either utilize prompt-based learning method to achieve encouraging results for few-shot tasks by determining an appropriate prompt template, or leverage unlabeled data by generating pseudo labels. However, the manual design of the prompt template requires not only labor-intensive but also expert knowledge. Roughly utilizing a semi-supervised method to assign pseudo-label to unlabeled data will suffer from noise accumulation. To tackle these problems, we propose a novel semi-supervised prompt learning framework with relation-aware gradient denoising for low-resource relation extraction(PromptRGD). Firstly, using learnable template words and virtual labels for prompt learning, we introduce entity type and relation prior knowledge into prompt template construction. Secondly, given the relation-aware gradient similarity between labeled and unlabeled data, PromptRGD generates a pseudo label and then improves the quality of pseudo labels on unlabeled samples in a self-training fashion. The main experimental results and a series of analyses prove the effectiveness of PromptRGD.**

***Index Terms*—low-resource relation extraction, semi-supervised prompt learning, self-training, gradient denoising**

I. Introduction

Relation extraction is a fundamental task to construct a knowledge graph and attracts a lot of attention. Due to its ability to extract structured triplets (subject, relation, object), it benefits many NLP high-level applications, e.g., Question answering, knowledge graph, and Information retrieval.

In the past few years, pre-trained language models(PLMs) such as GPT [1], BERT [2] and T5 [3] become a mainstream method for most natural language processing tasks. Compared with traditional methods, many neural network methods based PLM have achieved state-of-the-art result in many NLP tasks, including lots of RE benchmarks [4], [5]. However, since the learning objectives of the PLMs are often different from those of the actual tasks to be processed, it is necessary to further train extra classifier heads related to downstream task on the basis of PLM. Their performance is strongly reliant on expensive annotated data. This inspires a Low Resource Relation Extraction (LRE) problem. A series of research on prompt learning have emerged to solve this issue. This method first uses the prompt template to map the original input sentence to a final input sequence with at least one [MASK] token. Then, the model predicts a specific word for the [MASK] token. Finally, the predicted words and labels are mapped to get the final prediction results. Prompt learning matches the training objectives of pre-training and fine-tuning and has yielded superior results on few-shot task. Fig. 1 illustrates how pre-training, fine-tuning, and prompt learning respectively work.

Prompt learning mainly consists of two main methods: the discrete template construction method and the continuous template construction method [6]. The discrete method constructs the template of prompt in the form of natural language. One relies on expert experience and domain knowledge to design such prompt templates artificially. Another aims to generate discrete templates automatically through model generation or dictionary search [7], [8]. The continuous method constructs the template of prompt in the form of non-natural language. Specifically, this method initializes some special tokens randomly and then optimizes these tokens through training [9]. However, when applying the prompt learning method to improve the performance of relation extraction in low-resource scenarios, there are still several challenges to be addressed. On the one hand, designing an optimal prompt template and reasonable label word set for each relation category requires rich domain knowledge. Using search technology to automatically construct high-quality prompt templates and label word sets requires exponential computing costs to generate and verify [7], [8], [10], [11]. On the other hand, the model needs to fill multiple [MASK] tokens with corresponding label word, and finally integrate them to map to a single category. Furthermore, there is extensive semantic information among entity types and relation labels, which is critical. To reduce computation costs and reliance on expert experience, we inject prior knowledge directly or indirectly, such as rich entity type information, into prompt construction with learnable continuous tokens, rather than random initialization. With the continued optimization of training, the optimal prompt template can be obtained. On the other hand, to solve the problem of the multi-token mapping of prompt learning, we construct a trainable virtual prototype injected with prior knowledge, instead of transforming it into a multi-category and multi-label classification problem.

Meanwhile, many attempts are being made to strengthen the model's capabilities by diminishing the requirement for human supervision in relation extraction. To expand labeled data, the semi-supervised relation extraction approach utilizes a small amount of labeled data to train a model and then uses the model to assign pseudo-labels to unlabeled data. The model continues to train on the amplified data to improve overall performance. There are two major approaches to use unlabeled data: distant supervision method and self-training method. Distant Supervision leverages annotated triplets stored in external knowledge bases (KBs) as the supervision to generate a pseudo label for unlabeled samples [12], [13]. However, this approach presupposes that the same entitie pair have an identical relation. It ignores particular contexts, which makes the relation labels generated noisy and restricts their capacity to generalize. The self-training method [14] is a typical semi-supervised training method. It first learns a model with weak ability from the rare labeled data. Then it uses this model to allocate a pseudo-label to unlabeled data and adds the pseudo-label with high confidence to the labeled dataset. Finally, the model continues to train on the labeled dataset amplified to enhance its ability. Yet, the self-training strategy is plagued by the issue of gradual drift brought by noisy pseudo labels [15], [16]. GradLRE [17] alleviate the noise by drawing on the idea of trial and error to obtain the optimal solution from reinforcement learning to encourage pseudo-labeled data to be as close to labeled data as possible in the gradient descent direction. However, there are several challenges to this method: On the one hand, due to restricted annotations, pseudo labels initially generated by the model have more noise, which makes further optimization more challenging. On the other hand, this method urges pseudo-labeled data to be as close to labeled data as possible in the gradient direction based on the whole dataset, ignoring the gradient differences between relations. To alleviate the first challenge, we propose a prior knowledge-enhanced prompt learning method to improve performance when just a few labeled examples are available. Considering the differences between relations, we present a relation-aware gradient denoising method that, based on relation-aware gradient similarity, further reduces noise in pseudo label data.

In general, this paper has made the following contributions:

- We propose a novel semi-supervised prompt learning framework with relation-aware gradient denoising for LRE(PromptRGD). To the best of our knowledge, we are the first to apply the prompt learning method for the semi-supervised pseudo-label denoising problem.
- We propose a prior-knowledge-injected continuous prompt learning method when only a few labeled samples can be obtained. On the one hand, to reduce reliance on labor and expert experience, we adopt the design of continuous prompt and inject prior knowledge directly or indirectly, rather than random initialization. With the continued optimization of training, the optimal prompt template can be obtained. On the other hand, to solve the problem of the multi-token mapping of prompt learning, we construct a trainable virtual prototype, instead of transforming it into a multi-category and multi-label classification problem.
- We develop a relation-aware gradient denoising method that alleviates the issue of noise accumulation caused by the scarcity of labeled data, and further urges the model to produce high-quality pseudo-labels for unlabeled data by considering the gradient differences between relations.
- We show that our framework yields better performance over strong baselines. The main experimental results and a lot of analysis and discussion validate the effectiveness of our proposed approaches.

The remaining portions of this essay are organized as follows: In section II, we introduce our proposed framework in detail, including the prompt learning module and the relation-aware gradient denoising algorithm. We will present the main experimental results and a series of discussions in section III. In section IV, we describe the related work and compare them with our work. In section V, we summarize the main contributions, limitations and future work.

II. Framework

In this section, we will introduce our work in detail: a semi-supervised prompt learning framework with relation-aware gradient denoising for LRE. As shown in fig. 2, our proposed framework consists of two modules: Relation Extraction with Prompting Learning(REPL) and Pseudo Label Generation and Denoising(PLGD). To begin, the REPL maps labeled and unlabeled data into our designed knowledge-enhanced continuous prompt template. The model will then employ these mapped sentence sequences to learn and predict high-confidence pseudo-label for unlabeled samples. Unlike traditional self-training setting, the PLGD module finally optimize the quality of pseudo labels by minimizing the distance between the gradient of the pseudo-labeled data and the average gradient of the labeled data under the corresponding relation. We elucidate the details of REPL and PLGD.

A. Relation Extraction with Prompting Learning

We propose a knowledge-enhanced continuous prompt learning method that frees intensive labor and improves performance when only a few labeled samples can be obtained. We explain the details of the prompt learning method and how to construct prompt template as follows.

1) Prompt Learning: The prompt learning method aims to utilize better grammar and semantic knowledge stored in PLM by aligning the goal of the downstream task with the objectives of the pre-training stage. This method first needs a prompt template function $T(\cdot)$ to convert the original sentence x into new input sequence $x_{prompt} = T(x)$. Prompt template indicates the respective positions of the original input

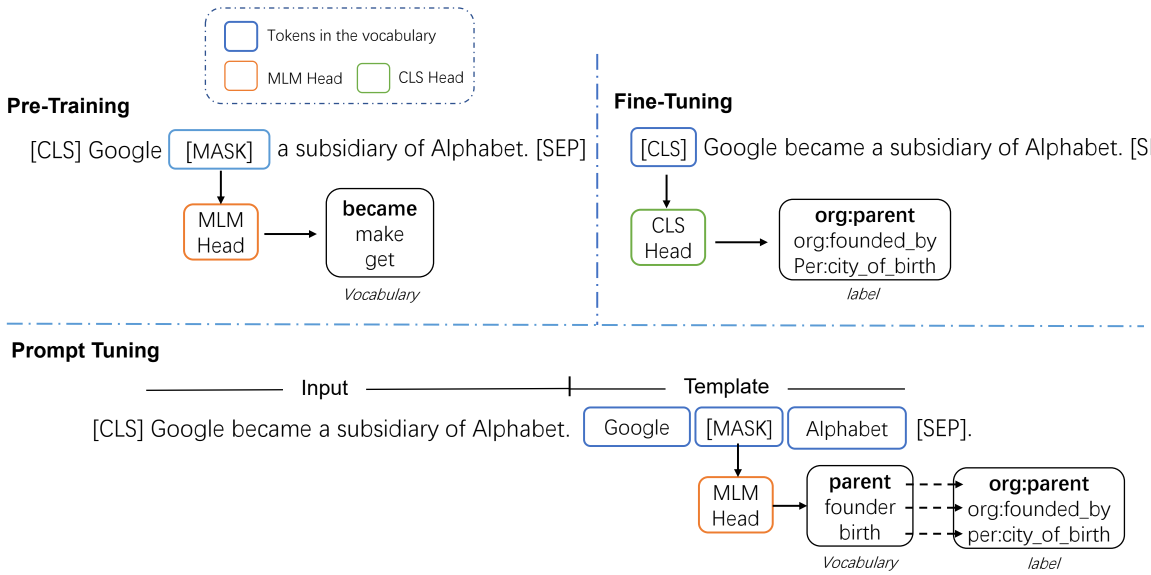

Fig. 1. The comparison of pre-training, fine-tuning, and prompt-learning.

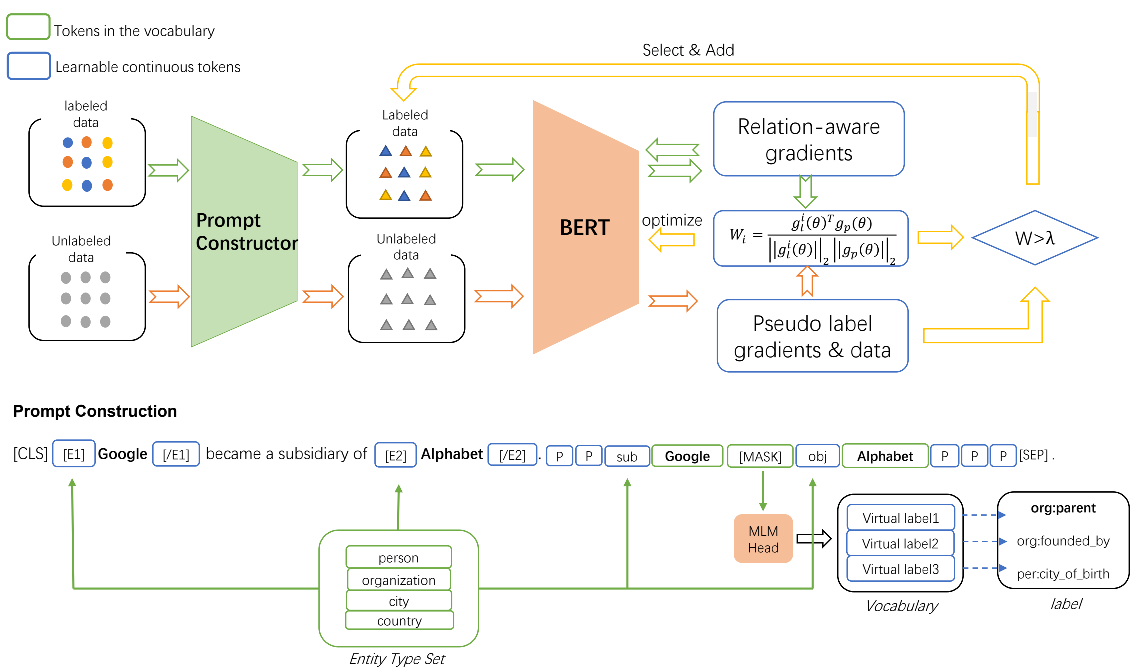

Fig. 2. Overview of the PromptRGD for Low Resource Relation Extraction.

sentence and additional information such as learnable tokens and special tokens. The converted x_{prompt} must contain at least one [MASK] token. Then the model M will select a reasonable word from the dictionary to match the [MASK] token according to the context of sequence, similar to cloze filling. We define the hidden state of the [MASK] token encoded by the model M as $\mathbf{h}_{[\mathbf{MASK}]}$, and the embedding of a word in M is $\mathbf{v}$. We then can calculate the probability of the word v filled into the masked position by the model M: $p([MASK] = v|x_{prompt}) = \frac{exp(\mathbf{v} \cdot \mathbf{h}_{[MASK]})}{\int_{v \in V} exp(\mathbf{v} \cdot \mathbf{h}_{[MASK]})}$. To get the final prediction results of each sample, the prompt learning method also has a mapping function $\phi : \mathcal{Y} \rightarrow \mathcal{V}$ that map a selected word to a relation category. Through this function, we can calculate the probability distribution of each category: $p(y|x) = p(\text{v}= \phi(y)|x_{prompt})$.

We take the relation extraction task as an example. First, we convert the original sentence x = "Google became a subsidiary of Alphabet." to x_{prompt} = "x. Google [MASK] Alphabet". through the prompt template function, which contains at least one MASK token. Encoded by the model M, we can get the hidden state $\mathbf{h}_{[\mathbf{MASK}]}$ of the MASK token and the probability distribution of all words in dictionary $\mathcal{V}$. We can select an appropriate word to fill the masked position according to the probability distribution of the dictionary. We map "per:parent" to "parent" label word and "org:founded by" to "founder" label word. According to whether the model fills "parent" or "founder" in the [MASK] position, we can identify the final classification result as "per: parent" or "org: founded by".

In summary, for prompt learning, according to the above parameter definitions, the learning goal is to maximize:

$$\frac{1}{|\mathcal{X}|} \sum_{x \in \mathcal{X}} \log p([MASK] = \phi(y_x)|T(x)) \tag{1}$$

2) Prompt Construction with Knowledge: Following the previous technique [18] with ENTITY/TYPE MARKER, we inject special symbols such as [E] and [/E] around entities to index the locations of entities. They are often employed in RE tasks when a sentence contains two entities: subject and object entity. However, apart from signaling places, these unique symbols can also hold entity-type information, providing the model with more distinguishable information. For example, when the entity type information is unknown, it is difficult to discern between the relation category "person:parent" and "organization:parent". We may acquire entity types given a specified relation label in our method, in addition to directly retrieving the type information via additional annotation. For example, for relation name per:country of birth", we can clearly see that the subject entity is person and the object entity is country. As a result, we employ learnable continuous tokens to comprehend entity-type information. Specifically, the learnable embeddings of virtual template words are initialized with entity-type embedding obtained directly or indirectly.

Previous research on prompt learning often created a one-to-one mapping between task labels and label words, which may not be effective for some tasks whose labels contain several tokens, like the relation extraction task. We believe that there is a continuous virtual label in the semantic space of PLM that can represent a class composed of multiple words, similar to a prototype in a prototypical network. In light of this, we translate the relation label into a virtual label that can be learned. Concretely, we use the weighted average of embeddings of each token in relation label tokenizations to initialize these virtual labels. Using the relation "org:parent " as an example, we may dissect it into tokens: "organization" and "parent"; Then, using the weighted average of each token's embeddings listed above to initialize a corresponding virtual label, which can then be optimized throughout training to obtain an optimal virtual label in the embedding space of PLMs.

B. Pseudo Label Generation and Optimization

This module seeks to utilize unlabeled data and assign fewer noise pseudo-labels to them, particularly when annotations are restricted. To accomplish this, we first obtain the gradient of each relation category under the labeled data and use it as the standard gradient of the corresponding relation. Then, we assume that the distance between its gradient and the gradient of the labeled data will be very small for the corresponding relation, When the pseudo-label sample is accurate enough. Based on this hypothesis, we develop a relation-aware gradient denoising method. Here, we go through this method in more depth. The whole gradient denoising workflow is illustrated in Algorithm 1.

We define (x, y) as labeled data from $\mathcal{D}_l$ and $(\hat{x}, \hat{y})$ as pseudo-labeled sample from $\mathcal{D}_p$. First, we use the only labeled sample converted by prompt template function to train a model with weak ability. Then, this model is applied to allocate pseudo-labels for unlabeled samples.The relation category with the maximum conditional probability will be assigned by the model as the pseudo-label of the corresponding sample: $\hat{y} = argmax(f_\theta(\hat{x}))$. f_θ is the network by parameterized.

We give the following definitions for $\mathcal{L}_l^i$ and $\mathcal{L}_p$. $\mathcal{L}_l^i$ represents the loss of labeled data under the i-th relation class. $\mathcal{L}_p$ refers to the loss of pseudo-labeled samples:

$$\mathcal{L}_l^i = \frac{1}{N_i} \sum_{n=1}^{N_i} loss\left(f_\theta(x^{(n)}), y^{(n)}\right) \tag{2}$$

$$\mathcal{L}_p = loss\left(f_\theta(\hat{x}), \hat{y}\right) \tag{3}$$

where $loss$ is the cross entropy loss function, i refers to the i-th relation class. For the sample, $f_\theta(x)$ yields a probability distribution covering all relation set, and y represent the target label. After obtaining two losses respectively through the loss function, we can calculate their gradients concerning model parameters. On labeled data, the gradient of the i-th relation category is defined as g_l^i, whereas on unlabeled data, it is defined as g_p:

$$g_l^{(i)}(\theta) = \nabla_\theta \mathcal{L}_l^i\left(x, y^i; \theta\right) \tag{4}$$

$$g_p(\theta) = \nabla_\theta \mathcal{L}_p\left(\hat{x}, \hat{y}; \theta\right) \tag{5}$$

Algorithm 1 Relation-aware Gradient Denoising in PromptRGD

Input: Labeled and Unlabeled dataset $\mathcal{D}_l$, $\mathcal{D}_p$, (x, y) as labeled data, $(\hat{x}, \hat{y})$ as pseudo-labeled data.
Output: Relation Extraction Network f_θ

1. Traing the model f_θ using limited labeled sample $\mathcal{D}_l$ and obtaining the expected relation-aware gradient $g_l(\theta)$.
for epoch in epochs **do**
  for batch in $\mathcal{D}_l$ **do**
    $\mathcal{L}_l^i = \frac{1}{N_i}\sum_{n=1}^{N_i} loss\left(f_\theta(x^{(n)}), y^{(n)}\right)$
    $g_l^{(i)}(\theta) = \nabla_\theta \mathcal{L}_l^i\left(x, y^i; \theta\right)$
    backward()
  end for
end for
2.Assigning pseudo-label $\hat{y}$ to $\hat{x}$ from 10% of $\mathcal{D}_p$ and obtaining the pseudo-label gradient $g_p(\theta)$.
$\mathcal{L}_p = loss\left(f_\theta(\hat{x}), \hat{y}\right)$
$g_p(\theta) = \nabla_\theta \mathcal{L}_p\left(\hat{x}, \hat{y}; \theta\right)$
3.Computing the gradient similarity W_i between $g_l^{(i)}(\theta)$ and $g_p(\theta)$ with respect to the i-th relation class.
$W_i = \frac{g_l^i(\theta)^T g_p(\theta)}{||g_l(\theta)||_2||g_p(\theta)||_2}$
4.Select and add high similarity sample $(\hat{x}, \hat{y})$ to the labeled dataset $\mathcal{D}_l$ and update the expected gradient.
if $W_i > \lambda$ **then**
  $\mathcal{D}_l \leftarrow \mathcal{D}_l \cup \mathcal{D}_p$
  $g_l^i \leftarrow \frac{1}{N_i+1}(N_i g_l^i + g_p)$
end if
5. Gradient denoising and encouraging the model to create high-quality pseudo label.
$\mathcal{L}(\theta) = \sum_{t=1}^{T} loss\left(f_\theta(\hat{x}^t), \hat{y}_i^t\right) \times W_i^t.$
backward()
6. Repeating 2 to 5 step above in new unlabeled data.
7. Finally, traing the model f_θ using augmented labeled data.

where ∇_θ Calculate the gradient of loss function concerning model parameters θ. After obtaining g_l^i and g_p, we measure the cosine similarity between them:

$$W_i = \frac{g_l^i(\theta)^T g_p(\theta)}{||g_l(\theta)||_2||g_p(\theta)||_2} \tag{6}$$

where W_i refers to the current pseudo label sample cosine similatiry with respect to the i-th relation class. When $W_i > \lambda$, λ is a threshold, we add sample $(\hat{x}, \hat{y}_i)$ to labeled dataset, and update the standard gradient that corresponds to the high-confidence relation:

$$\mathcal{D}_l \leftarrow \mathcal{D}_l \cup \mathcal{D}_p \tag{7}$$

$$g_l^i \leftarrow \frac{1}{N_i+1}(N_i g_l^i + g_p) \tag{8}$$

Finally, in order to eliminate the noise accumulation problem caused by limited labeled data and urge the model assigns high-quality pseudo labels, the model will be tuned on each batch using the following training objective:

$$\mathcal{L}(\theta) = \sum_{t=1}^{T} loss\left(f_\theta(\hat{x}^t), \hat{y}_i^t\right) \times W_i^t. \tag{9}$$

where W_i^t is the similarity weight, $loss$ is the cross entropy loss function, f_θ is the model by parameterized θ. We minimize $\mathcal{L}(\theta)$ by the gradient descent method to obtain optimal model parameters θ.

III. EXPERIMENT

We begin by introducing datasets, baselines, evaluation measures, and experimental settings, then describe the main experiment results. Finally, a full analyses and discussion is offered to show the benefits of PromptRGD.

A. Datasets

We follow [17] to conduct experiments on two public RE datasets, including SemEval benchmark dataset [24], and the TAC Relation Extraction Dataset(TACRED) [5]. The statistics for RE datasets are shown in Table II. The SemEval benchmark dataset is the industry standard for assessing relation extraction models. It includes training, validation, and test sets with 7,199, 800, and 1,864 relation mentions, respectively. It has a total of 19 relation categories, of which the relation category "no relation" accounts for 17.4%. TACRED is a sizable dataset produced by the crowd-sourcing of relation extraction from previous TAC KBP schema. It includes training, validation, and test sets with 75049, 25763, and 18659 relation mentions, respectively. It has a total of 42 relation categories, of which the relation category "no relation" accounts for 78.7%.

B. Baselines and Evaluation metrics

Considering that our PromptRGD can be widely applied to various encoders and models, we choose several common relation extraction models as base models, such as LSTM [19], PCNN [13], PRNN [5], BERT [2]. We only train base

TABLE I
The experimental results of different methods in various proportions of labeled data and fixed half of unlabeled data are compared. Methods located in the upper part of the table only use labeled samples.

Method/% Labeled Data	SemEval			TACRED		
	5%	10%	30%	3%	10%	15%
LSTM [19]	$22.65_{\pm 3.35}$	$32.87_{\pm 6.79}$	$63.87_{\pm 0.65}$	$28.68_{\pm 4.29}$	$46.79_{\pm 0.99}$	$49.42_{\pm 0.59}$
PCNN [13]	$41.82_{\pm 4.48}$	$51.34_{\pm 1.87}$	$63.72_{\pm 0.51}$	$40.02_{\pm 5.23}$	$50.35_{\pm 3.28}$	$52.50_{\pm 0.39}$
PRNN [5]	$55.34_{\pm 1.08}$	$62.63_{\pm 1.42}$	$69.02_{\pm 1.01}$	$39.11_{\pm 1.92}$	$52.23_{\pm 1.20}$	$54.55_{\pm 1.92}$
BERT [2]	$70.71_{\pm 1.24}$	$71.93_{\pm 0.99}$	$78.55_{\pm 0.87}$	$40.11_{\pm 3.88}$	$53.17_{\pm 1.67}$	$55.55_{\pm 0.82}$
$Self\_Training_{BERT}$ [14]	$71.34_{\pm 1.68}$	$74.25_{\pm 1.10}$	$81.71_{\pm 0.79}$	$42.11_{\pm 1.04}$	$54.17_{\pm 0.53}$	$56.52_{\pm 0.40}$
$MeanTeacher_{BERT}$ [20]	$70.05_{\pm 3.89}$	$73.37_{\pm 1.42}$	$80.61_{\pm 0.81}$	$44.34_{\pm 1.78}$	$53.08_{\pm 1.01}$	$53.79_{\pm 1.38}$
$DualRE-Pairwise_{BERT}$ [21]	$74.35_{\pm 2.63}$	$75.71_{\pm 1.39}$	$81.34_{\pm 0.74}$	$42.78_{\pm 1.89}$	$54.83_{\pm 0.95}$	$55.68_{\pm 1.21}$
$DualRE-Pointwise_{BERT}$ [21]	$74.02_{\pm 1.68}$	$77.11_{\pm 1.02}$	$82.91_{\pm 0.62}$	$43.71_{\pm 1.60}$	$56.28_{\pm 0.61}$	$57.72_{\pm 0.49}$
$MRefG_{BERT}$ [22]	$75.48_{\pm 1.34}$	$77.96_{\pm 0.90}$	$83.24_{\pm 0.71}$	$43.81_{\pm 1.44}$	$55.42_{\pm 1.40}$	$58.21_{\pm 0.71}$
$MetaSRE_{BERT}$ [23]	$78.33_{\pm 0.92}$	$80.09_{\pm 0.78}$	$84.81_{\pm 0.44}$	$46.16_{\pm 1.02}$	$56.95_{\pm 0.34}$	$58.94_{\pm 0.36}$
$GradLRE_{BERT}$ [17]	$79.65_{\pm 0.68}$	$81.69_{\pm 0.57}$	$85.52_{\pm 0.34}$	$47.37_{\pm 0.74}$	$58.20_{\pm 0.33}$	$59.93_{\pm 0.31}$
$PromptRGD_{BERT}$	$\mathbf{81.96_{\pm 0.30}}$	$\mathbf{83.91_{\pm 0.42}}$	$\mathbf{86.24_{\pm 0.28}}$	$\mathbf{47.45_{\pm 0.45}}$	$\mathbf{58.30_{\pm 0.28}}$	$59.87_{\pm 0.16}$
BERT w.gold labels	$84.64_{\pm 0.28}$	$85.40_{\pm 0.34}$	$87.08_{\pm 0.23}$	$62.93_{\pm 0.41}$	$63.66_{\pm 0.23}$	$64.69_{\pm 0.29}$

TABLE II
Statistics for RE datasets, including numbers of relations and instances and the proportion of "no relation" class

Dataset	Train	Val	Test	Rel	No Rel
SemEval	7199	800	1864	19	17.4%
TACRED	75049	25763	18659	42	78.7%

models on labeled data at initially, and the model that performs the best is chosen to serve as the primary encoder for our framework. Meanwhile, we select six representative semi-supervised relation extraction methods to compare with our method. For a fair comparison, all approaches employ the same encoder.

- The self-Training [14] method is a typical semi-supervised training method. It first learns a model with weak ability from the rare labeled data. Then it uses this model to allocate a pseudo-label to unlabeled data and adds the pseudo-label with high confidence to the labeled dataset. Finally, the model continues to train on the labeled dataset amplified to enhance its ability.
- Mean-Teacher [20] aims to enable multiple models to get a consistent standard answer based on the same sample. This method uses perturbation-based loss to make the output distribution of models as close as possible.
- DualRE [21] considers two tasks as dual tasks: obtaining relation between two entities and retrieving sentences according to entities and entity relation. It trains the model by combining the losses of two tasks.
- MRefG [22] first creates a reference graph using the entity and entity type information, part of speech information, entity relation information, and verb-noun reference information and then allocate a pseudo-tag to the unlabeled sample according to the reference graph.
- As an extra meta-objective, MetaSRE [23] uses meta-learning methods to synthesize the prediction results of different modules to allocate pseudo-labels to unlabeled sample.
- GradLRE [17] alleviate the noise by drawing on the idea of trial and error to obtain the optimal solution from reinforcement learning to encourage pseudo-labeled data to be as close to labeled data as possible in the gradient direction.

Finally, we present a special experiment setting: **BERT w. gold labels**. This model uses the same encoder and data quantity as other semi-supervised methods. The only disparity is that this model is trained on labeled data, and the unlabeled sample also has a gold label. This model provides a performance upper bound for all semi-supervised methods.

We chose the F1 score as the main metric for evaluation. It should be noted that the true predictions of "no relation" are discarded in the following.

C. Implementation Details

In this section, we will describe the data division method and parameter settings required for our experiment. First of all, strictly following the setting used in [17], we need to divide the entire public dataset into labeled and labeled dataset parts according to the category proportion, in which the gold label of unlabeled sample needs to be removed. Considering the large difference in data quantity among different datasets, we divide the Semeval dataset into three parts: 5%, 10%, and 30%, and the Tacred dataset into three parts: 3%, 10%, and 15%. We take 50% of the unlabeled data of the two datasets and then they will be assigned pseudo labels by optimizing. For all semi-supervised methods, we follow the setting by [17] for a fair comparison. After preliminary experiments on several common encoders, we chose the pre-trained BERT-Base model as the base model of PromptRGD. We set the maximum sequence length to 128. We use Adam optimizer with a 3e-5 learning rate and set batch size is 16 to train.

D. Main Results

The upper part of Table I shows the preliminary results of some common basic encoders on two public datasets when only limited labeled sample can be obtain. The second part

TABLE III
Ablation studies of PromptRGD on SemEval dataset

Method/% labeled data	5%	10%	30%
PromptRGD	$\mathbf{81.96}_{\pm\mathbf{0.30}}$	$\mathbf{83.91}_{\pm\mathbf{0.42}}$	$\mathbf{86.24}_{\pm\mathbf{0.28}}$
w/o pseudo-label optimation	$76.87_{\pm1.20}$	$83.18_{\pm0.53}$	$86.07_{\pm0.43}$
w/o prompt learning	$80.4_{\pm0.55}$	$82.8_{\pm0.40}$	$85.7_{\pm0.20}$

TABLE IV
F1 (%) comparisons on the SemEval datasets with different scenarios using prompt learning.

Method/% labeled data	5%	10%	30%
PromptRGD	$\mathbf{81.96}_{\pm\mathbf{0.3}}$	$\mathbf{83.91}_{\pm\mathbf{0.42}}$	$\mathbf{86.24}_{\pm\mathbf{0.28}}$
w/o prompt learning	$80.4_{\pm0.55}$	$82.8_{\pm0.4}$	$85.7_{\pm0.2}$
GradLRE	$79.65_{\pm0.68}$	$81.69_{\pm0.57}$	$85.52_{\pm0.34}$
GradLRE+prompt learning	$\mathbf{79.79}_{\pm\mathbf{0.44}}$	$\mathbf{82.64}_{\pm\mathbf{0.39}}$	$\mathbf{85.88}_{\pm\mathbf{0.32}}$
Bert w. gold	$84.64_{\pm0.28}$	$85.4_{\pm0.34}$	$87.08_{\pm0.23}$
Our appproach w. gold	$\mathbf{87.43}_{\pm\mathbf{0.12}}$	$\mathbf{87.41}_{\pm\mathbf{0.17}}$	$\mathbf{88.71}_{\pm\mathbf{0.19}}$

TABLE V
Predictions with/without RGD on SemEval, where red and cyan refer to entities respectively.

A child is told a lie for several years by their parents before he/she realizes that a Santa Claus does not exist
Label: **Product-Producer**
Prediction w/o RGD: **Message-Topic**
Prediction w. RGD: **Product-Producer**
It was in this background that the brigade of the guards was raised in 1949.
Label: **Member-Collection**
Prediction w/o RGD: **Other**
Prediction w. RGD: **Member-Collection**
She began to create a brew in her cauldron, that would take an entire year to prepare.
Label: **Content-Container**
Prediction w/o RGD: **Component-Whole**
Prediction w. RGD: **Content-Container**

shows the main experimental performance using a series of semi-supervised methods, including PromptRGD, on two public datasets when only rare label sample and half of unlabeled sample are used. Based on the upper part of Table, we see that the BERT [2] achieves the best results among all the basic encoders. To have an honest comparison, we choose the BERT [2] as the main model for all semi-supervised methods. Comparing the experimental results in the second half with the BERT model with only labeled data, we find that all methods using unlabeled data perform better. This indicates the utilization value of unlabeled data when there are only a few labeled data. We could also observe that GradLRE achieves the best performance which is considered the previous SOTA method. Our method surpasses all other semi-supervised learning methods. When compared with the best results GradLRE, we obtain an average of 1.75% higher score on the SemEval dataset and 0.04% higher score on the TACRED dataset under different proportions of labeled data settings. We assume that there are two reasons why there is no significant improvement on the TACRED dataset: On the one hand, the "no relation" relation class accounts for 78.7% of the TACRED dataset, which is extremely unbalanced. It implies the limitations of our approach in dealing with extremely unbalanced categories. On the other hand, [25] observed that 23.9% of TACRED labels are incorrect. In terms of less labeled data, PromptRGD can achieve better performance than the SOTA method. For instance, PromptRGD improves F1 performance by 2.31% with 5% percent labeled data on SemEval, demonstrating the power of PromptRGD when only limited labeled sample can be obtained. When the standard deviation is taken into account, our method outperforms all baselines.

E. Ablation Studies and Analysis

1) Ablation Studies: After our framework achieves better overall performance, we will conduct ablation experiments in this part to demonstrate the effectiveness of different modules in our framework. PromptRGD w/o pseudo-label optimization is the proposed framework without optimizing the pseudo label according to the relation-aware gradient similarity between labeled data and unlabeled data. PromptRGD w/o prompt learning is the proposed model without prompt learning module. Specifically, there is no mapping between the input sentence and prompt template, which is used to demonstrate the effectiveness of prompt learning. From the results of the ablation experiments in table III, we find that each module in the framework makes a positive contribution to the final performance, which demonstrates the effectiveness of each module. More quantitatively, performance on PromptRGD w/o pseudo-label optimization and w/o prompt learning are declined by 1.99% and 1.07% averaged over the Semeval dataset.

2) The adaptability of Prompt Learning: We conduct three groups of comparative experiments in different scenarios to show the effectiveness and adaptability of our proposed knowledge-enhanced learnable continuous prompt learning. The first group is equivalent to an ablation experiment about prompt learning. In the second group, the GradLRE w. prompt is the previous SOTA model with our prompt learning, which is used to compare with the initial GradLRE. At last, PromptRGD w. gold labels is only our prompt learning method when all data have gold labels. From experiment results in table IV, we found that performance on GradLRE w. prompt and PromptRGD w. gold labels are improved by 0.48% and 2.14% on F1 averaged over the Semeval dataset, the latter indicates the new upper bound of LRE models. In summary, three groups of comparative experiments demonstrate our proposed knowledge-enhanced learnable continuous prompt learning is effective and able to adapt to a variety of different scenarios and bring benefits when labeled data is scarce.

3) Analysis of alleviating the gradual drift problem: To utilize unlabeled sample, the naive self-training method directly uses a small number of labeled sample to learn a basic model and then applies this model to assign a pseudo label to unlabeled sample. Yet, such a strategy can easily lead to the problem of noise accumulation, and the quality of the pseudo-label evolves worse with the training. To alleviate this problem,

TABLE VI
Experiment results of analysis of alleviating the gradual drift problem on the SemEval datasets.

Method/% labeled data	5%	10%	30%
Self-Training	$71.34_{\pm 1.68}$	$74.25_{\pm 1.1}$	$81.71_{\pm 0.79}$
GradLRE	$79.65_{\pm 0.68}$	$81.69_{\pm 0.57}$	$85.52_{\pm 0.34}$
Relation-aware Gradient	$\mathbf{80.46_{\pm 0.55}}$	$\mathbf{82.8_{\pm 0.4}}$	$\mathbf{85.7_{\pm 0.2}}$

GradLRE [17] draws on the idea of trial and error to obtain the optimal solution from reinforcement learning to encourage pseudo-labeled data to be as close to labeled data as possible in the gradient descent direction. However, there are several challenges to this method: On the one hand, due to limited annotations, pseudo labels initially generated by the model have more noise, which makes subsequent optimization more difficult. On the other hand, this method urges pseudo-labeled sample to be as close to labeled sample as possible in the gradient descent direction based on the whole labeled dataset, which ignores the gradient differences between categories. In table VI, we conduct three comparative experiments based on the three methods above. The experiment result demonstrates that PromptRGD further alleviates the gradual drift problem.

We also make further case analyses on the SemEval dataset. The results are shown in Table V. Prediction w/o RGD tends to predict more "Other" categories, most likely because the "Other" class accounts for a large proportion. Secondly, Prediction w/o RGD often makes predictions based on the information of a single entity, such as associating the word "lie" with message, which may be due to the influence of noise. We also notice some cases where Prediction w. RGD performs better at distinguishing instances between similar relations such as Component-Whole and Content-Container.

IV. Related Work

A. Relation Extraction

Relation extraction is a fundamental task to construct a knowledge graph and attracts a lot of attention. Due to its ability to extract structured triplets (subject, relation, object), it benefits many NLP high-level applications. Traditional approaches involve template-based approaches, CNN/RNN-based methods, and graph-based methods. Pattern-based methods [26], [27] automatically construct pattern rules from grammatical elements. Feature-based methods [28]–[30] use feature engineering on entities and contexts before classification tasks. Methods based on CNN [31], RNN [32], and LSTM [19] introduce neural networks for relation extraction. Graph-based methods [33]–[35] construct entities graph for inferencing. With the advances in pre-training technology, applying PLMs as the main encoder of RE task [36]–[39] has become a mainstreamed trend. On this basis, [40] use entity type to constrain relation classification and achieve SOTA performance on TACRED [5]. Considering the fact that high-quality labeled data is rare, and the cost is high, a recent trend is few-shot settings. [41] construct the FewRel which is a few-shot relation extraction dataset. [42] focus on the application of snowball in Few-Shot Relation Learning.

B. Prompt Learning

With the emergence of GPT3 [43], prompt learning methods have become a novel paradigm for few-shot learning. [10], [11] uses pre-defined manually crafted templates in a few-shot learning setting. To reduce the reliance on labor-intensive and expert experience, automatically constructing template construction and label word collection based on language models and search technology have attracted extensive attention. [7] introduce a pre-trained model T5 to generate template tokens. To construct prompt templates and label word sets automatically, [8] proposed a strategy based on the gradient to enable language models to generate words needed by templates and label word sets. The prompt template is composed of not only words in the form of natural language but also continuous non-natural language tokens that can be learnable. Recently, continuous prompt learning methods have been widely studied. [44] propose Prefix-Tuning to optimize a continuous task-specific vector with few parameters. Furthermore, [9] propose P-tuning to insert trainable variables into the embedding input. For relation extraction, in order to solve the problem of many-to-one mapping when prompt learning is applied to relation extraction tasks, [45] proposes a model of combining several sub-prompts based on logical rules. [46] encode semantic knowledge by prompt template design.

In the design of our prompt learning method, we draw on the latest series of research on prompt learning. On the one hand, to reduce reliance on labor and expert experience, we adopt the design of continuous prompt and inject prior knowledge directly or indirectly, rather than random initialization. With the continued optimization of training, the optimal prompt template can be obtained. On the other hand, to solve the problem of the multi-token mapping of prompt learning, we construct a trainable virtual prototype, instead of transforming it into a multi-category and multi-label classification problem.

C. Semi-supervised Learning

Many attempts are being made to strengthen the model's performance by diminishing the requirement for human supervision in relation extraction. To expand labeled dataset, the semi-supervised relation extraction approach utilizes limited labeled sample to train a model and then uses this model to allocate pseudo-labels to unlabeled sample. The model continues to train on the amplified dataset to improve overall performance. There are two major approaches to use unlabeled data: distant supervision method and self-training method. Distant Supervision leverages annotated triplets stored in external knowledge bases (KBs) as the supervision to generate a pseudo label for unlabeled samples [12], [13]. However, this approach presupposes that the same entitie pair have an identical relation. It ignores particular contexts, which makes the relation labels generated noisy and restricts their capacity to generalize. The self-training method [14] is a typical semi-supervised training method. It first learns a model with weak ability from the limited labeled sample. Then it uses this model to allocate a pseudo-label to unlabeled sample and adds the pseudo-label sample with high confidence to the labeled dataset. Finally, the model continues to train on the labeled dataset amplified to enhance its ability. Yet, the self-training strategy is plagued by the issue of gradual drift brought by noisy pseudo labels [15], [16]. As an extra meta-objective, MetaSRE [23] uses meta-learning methods to synthesize the prediction results of different modules to allocate pseudo tags to unlabeled sample. GradLRE [17] alleviate the noise by drawing on the idea of trial and error to obtain the optimal solution from reinforcement learning to encourage pseudo-labeled data to be as close to labeled data as possible in the gradient direction.

V. Conclusion

This paper proposes a novel semi-supervised prompt learning framework with relation-aware gradient denoising for LRE (PromptRGD). To the best of our knowledge, we are the first to apply the prompt learning method for the semi-supervised pseudo-label denoising problem. PromptRGD is designed to free intensive labor and alleviates the bias when only a few labeled samples can be obtained. We evaluate our approach on two public datasets and the results demonstrate that PromptRGD outperforms strong baselines, especially with less training data. In future work, we plan to explore denoising methods in the case of extreme imbalance of data and extend our design to broader domains such as computer vision. We also look forward to improving the performance of LRE by exploring more effective low-resource learning methods.

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